

Examining the impact of socio-economic, regulatory, and demographic variables on BRICS banking efficiency: A Two-Stage Fuzzy Approach

Peter Wanke

COPPEAD Graduate Business School – Federal University of Rio de Janeiro. Rua Paschoal Lemme, 355. Cidade Universitária, Rio de Janeiro, Brazil. CEP 20520-120.

Email: peter@coppead.ufrj.br

Carlos P. Barros

ISEG – Lisbon School of Economics and Management, ULisboa and CESA – Research Centre on African, Asian and Latin American Studies; Rua Miguel Lupi, 20. 1249-078 Lisboa, Portugal

Email: cbarros@iseg.ulisboa

Md. Abul Kalam Azad

University of Malaya, Department of Applied Statistics, Faculty of Economics and Administration, 50603 Kuala Lumpur, Malaysia, Email: azadiiuc@gmail.com

Abstract: This paper presents an efficiency assessment of BRICS banking from 2010-2014, using an integrated two-stage fuzzy approach. BRICS is the acronym created by Goldman Sachs that accounts for Brazil, Russia, India, China, and South Africa. More precisely, Fuzzy TOPSIS is used to assess the relative efficiency of BRICS banking over the course of the years. In the second stage, fuzzy regressions based on different rule-based systems are used to predict the impact of socio-economic, regulatory, and demographic variables on banking efficiency. They are confronted with the bootstrapped truncated regressions with conditional α -levels proposed in Wanke, Barros, and Emrouznejad (2015). The results reveal that the country gross savings and the GINI index ratio have a significant, positive impact on efficiency levels. On the other hand, efficiency levels tend to be lower under high inflationary ratios. Specifically in terms of accuracy, fuzzy regressions outperformed by far the bootstrapped truncated regressions with conditional α -levels. Policy implications are derived.

Keywords: Banking; BRICS; Fuzzy TOPSIS; Fuzzy Regression; performance

1. Introduction

There is a growing number of studies conducted with different methods to assess performance in banking. These studies are often grouped into two main approaches, i.e., parametric and nonparametric (Berger & Humphrey, 1997; Fethi & Pasiouras, 2010; Porcelli,

2009). The most popular parametric method is known as the Stochastic Frontier Approach (SFA)(Sampaio, 2013), whereas, the most popular nonparametric methods is DEA (Data Envelopment Analysis) (Amsler, Lee, & Schmidt, 2009; Liu, Lu, Lu, & Lin, 2013a, 2013b; Paradi & Zhu, 2013; Zhou, Ang, & Poh, 2008). When put into perspective, however, nonparametric methods are widely used in banking efficiency in various countries and regions around the globe (Liu et al., 2013b; Paradi & Zhu, 2013; Porcelli, 2009; Wanke, Azad, & Barros, 2016; Wanke, Azad, Barros, & Hadi-Vencheh, 2015). Thus far, to the best of our knowledge, multi-criteria decision-making methods, such as TOPSIS, have not been used to assess efficiency in banking at the level of economic blocks (Behzadian, Khanmohammadi Otaghsara, Yazdani, & Ignatius, 2012).

Empirical research on bank efficiency has been intensive over the last decades (Behzadian et al., 2012; Berger & Humphrey, 1997; Kahraman, Onar, & Oztaysi, 2015; Liu et al., 2013a, 2013b; Sampaio, 2013). Data envelopment analysis (DEA) is now in common among the bank efficiency studies (Paradi & Zhu, 2013). In the traditional DEA, efficiency is calculated using the ex post information (Berger & Humphrey, 1997; Charnes, Cooper, & Rhodes, 1978). However, Battese and Rao (2002) revealed that examining efficiency with DEA exhibits better discrimination if ex ante information is considered through meta-frontier analysis. Literature on DEA also suggests that most of the banking studies have considered developed economies as the focus group.

This paper, therefore, fills a literature gap by analyzing and exploring the sources of efficiency in BRICS banking by using a two-stage fuzzy approach. It innovates in this context not only by applying Fuzzy TOPSIS to assess efficiency levels but also by adopting different fuzzy regressions to assess the impact of demographic, regulatory, and socio-economic variables on these fuzzy efficiency levels. These results are further compared to those obtained using bootstrapped truncated regressions with conditional α -levels, as proposed in Wanke, Barros, and Emrouznejad (2015). This combination of fuzzy and probabilistic approaches also represents a contribution to the emerging literature on possible analytical venues within the ambit of 2-Dimensional Fuzzy Monte Carlo Analysis (2D FMCA).

The major contributions of this present research are the following. First, this research evaluates the evolution of the relative efficiency in BRICS banking, adding to the scarce literature on banking efficiency at the level of economics blocks (Cheng, Gutierrez, Mahajan, Shachmurove, & Shahrokhi, 2007; Choudhury & Ashraf, 2016; Zhang, Jiang, Qu, & Wang,

2013). Second, this research uses a Fuzzy TOPSIS method to compute efficiency based on Triangular Fuzzy Numbers (TFN) to assess vagueness in banking inputs and outputs. As a matter of fact, the banking outputs and inputs present different forms of uncertainty within their relationships. For example, credit granting is an input embedded in fuzziness because of the ex-ante risks associated with non-performing loans (Demirgüç-Kunt & Huizinga, 1999). On the other hand, the value of banking production, which is not a constant number, changes because of the economic conditions of the country. To evaluate banking efficiency more realistically and accurately, this study employs the fuzzy TOPSIS model with data specified in bounded forms to capture vagueness and uncertainty to some extent. Third, this research also expands the existing literature due to the use of fuzzy regressions and different rule-based systems to predict and interpret the role of major contextual variables in achieving higher levels of efficiency. Fourth, our analysis covers the period from 2010 to 2014, after the world financial crisis of 2007-2008. Finally, our analysis is based on a representative group of emerging economies.

Results presented in this research constitute a contribution to the growing literature on banking performance at the level of economic blocks. These results are consistent with and extend the findings of recently conducted studies, thus adding to the body of knowledge. The efficiencies scores were calculated using Fuzzy TOPSIS using a meta-frontier (O'Donnell, Rao, & Battese, 2008). Results reveal that efficiency levels among the banks tend to be quite similar between countries, although it is interesting to note that, efficiency in Chinese banks is upwards biased; Brazilian banks tend to present higher dispersion. Earlier, Wanke, Barros, and Faria (2015) examined Brazilian bank efficiency using dynamic slack based model and their results indicate that higher inefficiency and slacks are present in small public and national banks. Thus, a higher dispersion of efficiency results among different types of Brazilian banks.

Since O'Neill (2001) enlighten the world with BRICS, an emerging volume of literature has been examining the potentials of BRICS from a number of perspectives- market opportunities (Cheng et al., 2007; Mobarek & Fiorante, 2014; O'Neill, 2011); raising power (Jacobs & Van Rossem, 2014; Laïdi, 2012); competitive input of factors (Luminata et al., 2014); strong reserve of foreign currencies (Luminata et al., 2014); etc. Researchers like Jacobs and Van Rossem (2014) and O'Neill (2011) argued that the minimal effect from global crisis in 2008 on these countries along with growth potential of last decades compared to G7 or G20 (c.f. Luna (2016)) have shifted world focus from developed economies to emerging economies, especially regional integrations i.e., BRICS. On top of these, the access amount of foreign currency

reserves and high investment rates among these countries have been prioritizing BRICS over the developed countries' integration (Luminita et al., 2014).

The remainder of the paper is organized as follows: Section 2 covers the literature review; and Section 3 presents the methodology. The empirical results are presented and discussed in terms of policy implications in Section 4. Conclusions follow in Section 5.

2. Literature Review

In an earlier survey on efficiency of financial studies, Berger and Humphrey (1997) examined 130 international efficiency studies and revealed that five different approaches have mostly applied in these studies. Among the nonparametric approaches data envelopment analysis (DEA) and free disposal hull (FDH) are the major ones. Alternatively, the most popular models of parametric approach are stochastic frontier approach (SFA), distribution free approach (DFA) and thick frontier approach (TFA). Later on, the most recent survey on DEA (Liu et al., 2013b) and SFA (Amsler et al., 2009) reveal that the major application area of these efficiency models are banks and other financial institutions. These efficiency models are appropriate on specific circumstances and thus combined with both advantages and limitations (Berger & Humphrey, 1997).

Most recently, Kahraman et al. (2015) examined the application of different multicriteria decision making (MCDM) approaches in efficiency estimation and reveal that considering real-world decision making problems, the earlier frontier approaches have over simplified of an actual problem and failed to experiment a problems which is complex and ill-structured in nature. Moreover, the bank level data we often use for examining efficiency are vague and incomplete. For such an environment fuzzy MCDM is the best alternative (Dubois, 1980; Kahraman et al., 2015). The latest development and major applications of fuzzy MCDM can be found in Carlsson and Fullér (1996); Teh et al. (2013) and Kahraman et al. (2015).

3. Methodology

3.1. The data

The data on BRICS banking were obtained from different sources such as Bankscope database & Worldbank dataset and encompassed the period from 2010 to 2014. The negative criteria and the positive criteria considered observed not only those commonly found in the literature review but also the availability of data. Within the ambit of the Fuzzy TOPSIS model

depicted in Section 4.3, the negative criteria (input variables) included less reserves for impaired loans / NPLs, total capital ratio, Tier 1 ratio, loan loss res / gross loans, loan loss prov / net int rev, fixed assets, non-earning assets, equity, total liabilities & equity, loan loss reserves (memo), liquid assets (memo), overheads, loan loss provisions and tax. The positive criteria (output variables) included equity / tot assets, equity / net loans, equity / short term funding, equity / liabilities, net interest margin, return on avg. assets (ROAA), return on avg. equity (ROAE), cost to income ratio, recurring earning power, interbank ratio, net loans / tot assets, loans, total earning assets, total assets, deposits & short term funding, other (non-interest bearing), net interest revenue, other operating income, profit before tax and net income. Their descriptive statistics are presented in Table 1.

Table 1. Descriptive statistics for the inputs, outputs and the contextual variables

Variables		Min	Max	Mean	SD	CV
Negative criteria	Less Reserves for Impaired Loans / NPLs	0.000	15.425	13.212	0.360	0.027
	Total Capital Ratio	0.000	4.426	3.907	0.254	0.065
	Tier 1 Ratio	0.000	4.323	3.765	0.171	0.045
	Loan Loss Res / Gross Loans	0.000	3.598	1.668	0.638	0.382
	Loan Loss Prov / Net Int Rev	0.000	6.136	5.822	0.148	0.025
	Fixed Assets	0.000	15.227	4.202	1.940	0.462
	Non-Earning Assets	0.000	15.750	13.365	0.414	0.031
	Equity	0.000	14.993	6.916	1.606	0.232
	Total Liabilities & Equity	0.000	17.702	9.094	1.885	0.207
	Loan Loss Reserves (Memo)	0.000	12.917	11.339	0.323	0.028
	Liquid Assets (Memo)	0.000	16.858	11.711	0.586	0.050
	Overheads	0.000	12.904	4.942	1.852	0.375
	Loan Loss Provisions	0.000	10.629	10.375	0.245	0.024
	Tax	0.000	11.854	7.992	0.492	0.062
Positive criteria	Equity / Tot Assets	0.000	4.801	3.553	0.287	0.081
	Equity / Net Loans	0.000	7.565	6.883	0.177	0.026
	Equity / Cust & Short Term Funding	0.000	6.561	4.132	0.426	0.103
	Equity / Liabilities	0.000	6.703	3.523	0.450	0.128
	Net Interest Margin	0.000	4.705	3.731	0.143	0.038
	Return On Avg Assets (ROAA)	0.000	4.188	3.557	0.127	0.036
	Return On Avg Equity (ROAE)	0.000	7.038	5.985	0.154	0.026
	Cost To Income Ratio	0.000	6.432	3.983	0.516	0.130
	Recurring Earning Power	0.000	4.426	3.391	0.153	0.045
	Interbank Ratio	0.000	6.937	5.197	0.777	0.149
	Net Loans / Tot Assets	0.000	4.929	3.956	0.462	0.117
	Loans	0.000	16.737	9.465	1.332	0.141
	Total Earning Assets	0.000	17.555	10.958	1.016	0.093
	Total Assets	0.000	17.702	9.094	1.885	0.207
	Deposits & Short term funding	0.000	17.596	11.301	0.939	0.083
	Other (Non-Interest bearing)	0.000	14.025	6.312	1.495	0.237
	Net Interest Revenue	0.000	13.883	6.033	1.559	0.258
	Other Operating Income	0.000	13.123	9.242	0.404	0.044
	Profit before Tax	0.000	13.535	10.000	0.450	0.045
	Net Income	0.000	13.130	9.986	0.405	0.041
Socio-economic and demographic variables	Bank capital to assets ratio (%)	6.100	12.896	8.425	1.999	0.237
	Bank liquid reserves to bank assets ratio (%)	3.453	27.671	17.379	9.567	0.550
	GDP growth (annual %)	0.103	10.632	4.874	3.067	0.629
	GDP per capita (constant 2005 US\$)	1010.31	6922.803	4607.503	2130.120	0.462
	GDP per capita growth (annual %)	-1.139	10.099	3.900	3.218	0.825
	Gross savings (% of GDP)	14.351	50.603	28.452	12.587	0.442
	Inflation, GDP deflator (annual %)	0.853	15.914	6.869	3.218	0.468
	GINI index (World Bank estimate)	33.850	63.380	46.300	10.013	0.216

3.2. Background on TFNs and TFMs

TFNs (triangular fuzzy numbers) are commonly found in Fuzzy TOPSIS uses (Behzadian et al., 2012; Syamal & Pal, 2007). On the other hand, it is well known that the matrix formulation of a mathematical formula provides additional ease of handling for a specific problem (Syamal & Pal, 2007). This being the case, introduced and derived the properties of TFMs (triangular fuzzy matrices) in a seminal work. On the other hand, fuzzy matrices were introduced for the first time by Thomason (1977), who discussed the convergence of powers of fuzzy matrices. In this line, several authors have presented a number of results on the convergence of the power sequence of fuzzy matrices (Hashimoto, 1983; Kandel, 1986). Broadly speaking, a Triangular Fuzzy Number " a " may be represented by $(a - \alpha, a, a + \beta)$, where α and β represents, respectively, the spreads to the left and to the right from a . Alternatively, fuzzy numbers of these kind can be expressed such as $\langle a, \alpha, \beta \rangle$. If M is a TFN expressed as $M = \langle a, \alpha, \beta \rangle$ its membership function is given by:

$$\mu_M(x) = \begin{cases} 0 & \text{for } x \leq m - \alpha \\ 1 - \frac{m - x}{\alpha} & \text{for } m - \alpha < x < m \\ 1 & \text{for } x = m \\ 1 - \frac{x - m}{\beta} & \text{for } m < x < m + \beta \\ 0 & \text{for } x \geq m + \beta \end{cases} \quad (1)$$

The membership function equals 1 when x reaches the mean value, m . Besides, considering α and β to be, respectively, the spreads to the left and to the right of the TFN M , it is possible to affirm that this number is symmetrical around the mean if both spreads assume the same value, that is, if $\alpha = \beta$.

Several researches have attempted to define the arithmetic operations of TFNs over the course of time. They were pioneered by Dubois and Prade (1998), who introduced the definitions of their arithmetic operations. Consider $M = \langle m, \alpha, \beta \rangle$ and $N = \langle n, \gamma, \delta \rangle$ to be two Triangular Fuzzy Numbers. Their arithmetic operations are defined as it follows (Syamal & Pal, 2007):

Addition: $M + N = \langle m + n, \alpha + \gamma, \beta + \delta \rangle$.

Scalar multiplication: If λ is scalar, $\lambda M = \langle \lambda m, \lambda \alpha, \lambda \beta \rangle$ when $\lambda \geq 0$.

Otherwise $\lambda M = \langle \lambda m, -\lambda \beta, -\lambda \alpha \rangle$ when $\lambda \leq 0$. Particularly, $-M = \langle -m, \beta, \alpha \rangle$

Subtraction: $M - N = \langle m, \alpha, \beta \rangle - \langle n, \gamma, \delta \rangle = \langle m - n, \alpha + \delta, \beta + \gamma \rangle$.

Given two triangular Fuzzy Numbers, M and N , their addition, subtraction, and scalar multiplication, i.e., $M + N$, $M - N$ and λM are all TFNs.

Multiplication: One may show that the membership function shape of $M \cdot N$ is not necessarily triangular. However, if the spreads of M and N are sufficiently small compared to their mean values m and n , then this shape follows a triangular form.

A robust decision rule is given next : (Syamal & Pal, 2007)

(a) When $M \geq 0$ and $N \geq 0$ ($M \geq 0$, if $m \geq 0$)

$$M \cdot N = \langle m, \alpha, \beta \rangle \cdot \langle n, \gamma, \delta \rangle \simeq \langle mn, m\gamma + n\alpha, m\delta + n\beta \rangle.$$

(b) When $M \leq 0$ and $N \geq 0$

$$M \cdot N = \langle m, \alpha, \beta \rangle \cdot \langle n, \gamma, \delta \rangle \simeq \langle mn, n\alpha - m\delta, n\beta - m\gamma \rangle$$

(c) When $M \leq 0$ and $N \leq 0$

$$M \cdot N = \langle m, \alpha, \beta \rangle \cdot \langle n, \gamma, \delta \rangle \simeq \langle mn, -n\beta - m\gamma, -n\alpha - m\delta \rangle$$

When spreads are not small compared with mean values, a better approximation is given next:

$$\langle m, \alpha, \beta \rangle \cdot \langle n, \gamma, \delta \rangle \simeq \langle mn, m\gamma + n\alpha - \alpha\gamma, m\delta + n\beta + \beta\delta \rangle \text{ for } M > 0, N > 0.$$

On the other hand, a Triangular Fuzzy Matrix (TFM) of order $m \times n$ can be given as $A = (a_{ij})_{m \times n}$, where $a_{ij} = \langle m_{ij}, \alpha_{ij}, \beta_{ij} \rangle$ is the ij th elements of A , m_{ij} is the mean value of a_{ij} and α_{ij}, β_{ij} are, respectively, the left and right spreads of a_{ij} . Likewise classical matrix algebra, let us consider the following operations involving TFMs, given that $A = (a_{ij})$ and $B = (b_{ij})$ are two TFMs of same order. In such cases, the following relationships are observed (Syamal & Pal, 2007).

$$(i) A + B = (a_{ij} + b_{ij})$$

$$(ii) A - B = (a_{ij} - b_{ij}),$$

$$(iii) \text{ For } A = (a_{ij})_{m \times n} \quad \text{and} \quad B = (b_{ij})_{n \times p}, \quad C = A \cdot B = (c_{ij})_{m \times p}$$

$$c_{ij} = \sum_{k=1}^n a_{ik} \cdot b_{kj}, \quad i = 1, 2, \dots, m \text{ and } j = 1, 2, \dots, p$$

$$(iv) A' = (a_{ji}) \text{ (the transpose of } A)$$

$$(v) k \cdot A = (ka_{ij}), \text{ where } k \text{ is a scalar.}$$

3.3. Fuzzy TOPSIS

The fuzzy TOPSIS method was proposed by Chen (2000). The difference in relation to the primary version of this method is in expressing assessments and/or criteria's weights in the form of triangular fuzzy numbers. An example of applying this method can be found, among others, in the studies of: S.-H. Chang and Tseng (2008), Uyun and Riadi (2013), Madi and Tap (2011), Yayla, Yildiz, and Ozbek (2012), and Kia, Danaei, and Oroei (2014).

Let us assume that a certain set of objects $A = (A_i \mid i = 1, \dots, n)$ and a set of criteria $C = (C_j \mid j = 1, \dots, m)$, where $\tilde{X} = \{\tilde{x}_{ij} \mid i = 1, \dots, n, j = 1, \dots, m\}$ stand for a set of fuzzy evaluation criterion and $\tilde{W} = \{\tilde{w}_j \mid j = 1, \dots, m\}$ a set of fuzzy weights. The linear ordering of objects with the application of the fuzzy TOPSIS method with the above outlined assumptions requires the accomplishment of the following steps (Chen, 2000):

Step 1. Calculation of normalized fuzzy evaluation criteria:

$$\tilde{z}_{ij} = \frac{\tilde{x}_{ij}}{\sqrt{\sum_{i=1}^n \tilde{x}_{ij}^2}}, \quad i = 1, \dots, n; \quad j = 1, \dots, m. \quad (2)$$

Step 2. Calculation of weighted normalized fuzzy evaluation criteria:

$$\tilde{v}_{ij} = \tilde{w}_j \tilde{z}_{ij} \quad (3)$$

Step 3. Appointing positive-ideal solution A^+ and negative-ideal solution A^- development:

$$\tilde{A}^+ = \{\tilde{v}_1^+, \tilde{v}_2^+, \dots, \tilde{v}_m^+\} = ((\max_i \tilde{v}_{ij} \mid j \in J_1), (\min_i \tilde{v}_{ij} \mid j \in J_2) \mid i = 1, \dots, n) \quad (4)$$

$$\tilde{A}^- = \{\tilde{v}_1^-, \tilde{v}_2^-, \dots, \tilde{v}_m^-\} = ((\min_i \tilde{v}_{ij} \mid j \in J_1), (\max_i \tilde{v}_{ij} \mid j \in J_2) \mid i = 1, \dots, n) \quad (5)$$

where J_1 and J_2 are respectively the benefit criterion and the cost criterion.

Step 4. Calculation for each object of a distance from positive-ideal solution d_i^+ and negative-ideal solution d_i^- (in the original work it is a Euclidean distance).

Step 5. Calculation of a synthetic measure:

$$CC_i^+ = \frac{d_i^-}{d_i^+ + d_i^-}, \quad i = (1, \dots, n). \quad (6)$$

Measure values (6) are normalized in an interval $< 0;1 >$. The smaller the distance of an object from a positive-ideal solution and the bigger from a negative-ideal solution, the closer the value of a synthetic measure is to cohesion.

Step 6. Establishing the objects' ranking. The best object has the biggest value of a synthetic measure.

In summary, the fuzzy TOPSIS method assumes in the first step of procedure, the normalization of fuzzy numbers according to the formula of linear scale transformation. According to the second step of the fuzzy TOPSIS method's procedure, the weights of particular criteria can be expressed in the form of TFN. In this research, for the sake of simplicity, the same system of weights has been assumed for all variables, therefore the parameters' values of fuzzy numbers representing weights are the same and come to 1. Besides, R codes provided in Jefmański and Dudek (2015) were used in this research for computation of the fuzzy TOPSIS scores.

3.4. Fuzzy Regression

Fuzzy regression was introduced by Asai (1982) to model situations in which the practitioner cannot accurately measure the dependent variable. As long as traditional statistical regression models can only fit crisp data, fuzzy linear regression models can be used to fit both fuzzy and crisp data (O. Y.-H. Chang & Ayyub, 2001). For such data, fuzzy set theory provides a means for modelling linguistic variables utilizing membership functions. In contrast to the traditional statistical regression models which are based on probability theory, fuzzy regression is based simultaneously on possibility theory (Dubois & Prade, 1998) and fuzzy set theory (Zadeh, 1965).

3.5. Bootstrapped Truncated Regression with conditional α -levels

Wanke, Barros, and Emrouznejad (2015) departed from the approach of Simar and Wilson (2007) and proposed conditional bootstrapped truncated regression to analyze the crisp values derived from fuzzy efficiency models – where scores range between 0 and 1 - based on α -levels. The following conditional modeling was proposed:

$$\theta_j | \alpha = k + Z_j \delta + F_j \gamma + \varepsilon_j, j = 1, \dots, n \quad (7)$$

The modeling can be understood as the first-order approximation of the unknown true relationship. In eq. (7), α is a real value bounded between 0 and 1 and represents the α -level of

the membership function for the efficiency score, k is the constant term, ε_j is statistical noise, F_j is vector of dummy variables that represent the fixed effects for the type of the fuzzy models used, whenever different models are used, and Z_j is a vector of the contextual variables for observation j that is expected to be related to the observation's efficiency score, θ_j , taken as a crisp value.

Specifically, noting that the distribution of ε_j is restricted by the condition $\varepsilon_j \geq 1 - k - Z_j\delta - F_j\gamma$ (since both sides of (7) are bounded by unit), the steps proposed in Simar and Wilson (2007) were followed in Wanke, Barros, and Emrouznejad (2015), and it was assumed that this distribution is truncated normal with zero mean (before truncation), unknown variance, and (left) truncation point determined by this very condition. Furthermore, replacing the true but unobserved regressand in (6), θ_j , by the respective fuzzy efficiency estimate, $\bar{\theta}_j$, the conditional econometric model formally becomes:

$$\bar{\theta}_j | \alpha \approx k + Z_j\delta + F_j\gamma + \varepsilon_j, j = 1, \dots, n, \quad (8)$$

where

$$\varepsilon_j \sim N(0, \sigma_\varepsilon^2), \text{ so that } \varepsilon_j \geq 1 - k - Z_j\delta - F_j\gamma, j = 1, \dots, n, \quad (9)$$

which is evaluated via maximal likelihood estimation as regards $(\delta, \sigma_\varepsilon^2)$ obtained from the data. It should be noted that in this research only one type of fuzzy TOPSIS model was used, thus implying the discard of vector F_j . Besides, the respective computations for the parametric bootstrap for this conditional regression were carried out with R codes developed by Wanke, Barros, and Emrouznejad (2015).

3.6. Rule-based systems in fuzzy regression

Fuzzy set theory was proposed by Zadeh (1965), as an extension of the classical set theory to model sets whose elements have degrees of membership. So, instead of just having two values: member or non-member, fuzzy sets allow for degrees of set membership, defined by a value between zero and one. A degree of one means that an object is a member of the set, a value of zero means it is not a member, and a value somewhere in-between shows a partial degree of membership. During the last forty years, scientific research has been growing steadily and the

available literature is vast (Riza, Bergmeir, Herrera, & Benítez Sánchez, 2015). A lot of monographs provide comprehensive explanations about fuzzy theory and its techniques, for example in Klir and Yuan (1995) and Pedrycz and Gomide (1998). One of the most fruitful developments of fuzzy set theory are fuzzy-rule based systems (FRBSs).

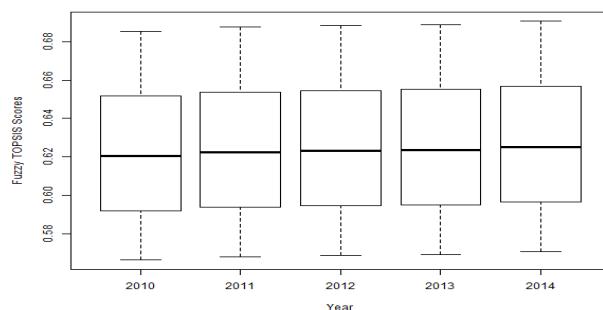
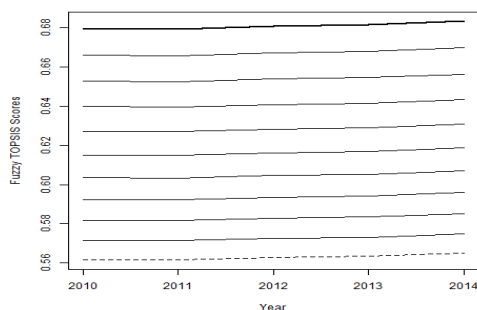
The parametrization of the respective models in the **frbs** R package (Riza et al., 2015) is given in Table 2.

Table 2. Parameters for the FRBS.

Methods	Parameters
ANFIS	num.labels = 5, max.iter = 300, step.size = 0.01, type.mf = 3
HYFIS	num.labels = 5, max.iter = 200, step.size = 0.01
SBC	r.a = 0.3, eps.high = 0.5, eps.low = 0.15
DENFIS	Dthr = 0.15, max.iter = 5000, step.size = 0.01, d = 2 "
FIR.DM	num.labels = 5, max.iter = 1000, step.size = 0.01
FS.HGD	num.labels = 5, max.iter = 100, step.size = 0.01, alpha.heuristic = 1
GFS.FR.MOGUL	persen_cross = 0.9, max.iter = 300, max.gen = 200, max.tune = 500, persen_mutant = 0.3, epsilon = 0.95
WM	num.labels = 15, type.mf = 3, type.defuz = 1, type.tnorm = 1, type.snorm = 1

4. Results and Discussion

It is interesting to note that, in a quite analogous way to what happen with bootstrapped estimates in frontier methods, fuzzy TOPSIS efficiencies are higher when there is no fuzziness at all (α -level = 1, represented by the solid bold line in Fig. 1 on the left). On the other hand, fuzzy TOPSIS efficiencies systematically decay with the value of α -level (thus representing increased fuzziness). Their minimal values are obtained when α -level = 0 and fuzziness is maximal. The dashed fine line in Fig. 1 on the left represents this. Under bootstrapping, the newer efficiency estimates computed statistically tend to be lower than the original ones.



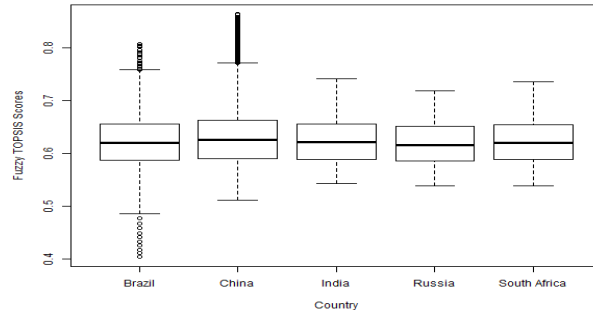


Fig. 1. Fuzzy estimates of BRICS banking efficiency levels.

Results for the conditional bootstrapped truncated regression performed on different α -levels reflect the impact of socio-economic, regulatory, and demographic variables on banking efficiency within the ambit of BRICS under different levels of fuzziness. They are presented in Fig. 2. From a quick inspection, several conclusions can be drawn. A solid line marks the zero in each graph, thus indicating whether a contextual variable is significant or not for a given value of α . Two contextual variables are not statistically significant, regardless of the uncertainty level in inputs and outputs: GDP per capita growth and bank capital.

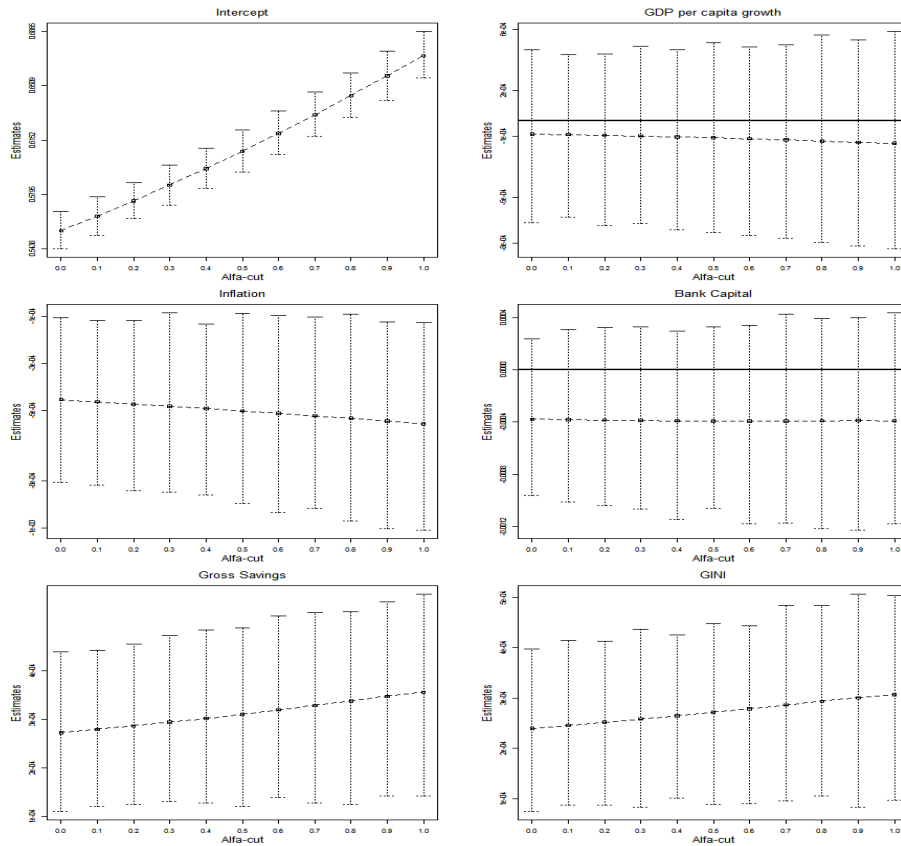


Fig. 2. Coefficient estimates for the conditional bootstrapped truncated regression.

However, the reminder contextual variables are significant regardless of the α -values: country gross saving, inflation ratio, and GINI index. They all present a positive impact on bank efficiency in the BRICS, thus suggesting that, although gross saving may be a prerequisite for higher productivity in banking systems, the impact of the inflation ratio in the loss of productivity cannot be neglected. Besides, there is an interesting counterbalance of social welfare and capital accumulation in banking productivity in BRICS. Efficiency tends to increase with higher GINI levels, which reflects income inequality and capital accumulation to some extent also, exerts a positive effect on efficiency.

It is also interesting to note that the sign of the significant impact did not depend upon the fuzziness level that the input and output data are subjected to. This suggests that uncertainty and randomness do not interact in the input/output level. This lack of ambiguity -which is not so often found in fuzzy systems applied to efficiency measurement (Wanke, Barros, & Emrouznejad, 2015; Wanke, Barros, & Nwaogbe, 2016) - opens the room for researchers and practitioners to investigate further the actual sources of efficiency using different rule-based systems within the ambit of fuzzy regressions. The inherent statistical limitations of fuzzy regressions previously discussed in Section 4.4 should be, of course, observed.

Bootstrapping was also used to build confidence intervals for the log-likelihood measurements for the conditional regression models considering different α -values. These results are presented in Fig. 3. Although one cannot claim that these log-likelihoods are significantly different, since their error bars overlap, it is interesting to note how the regression's performance is affected by the extreme values of uncertainty in the measurement of inputs and outputs. The best fit to the data was verified under $\alpha = 0$, that is, when fuzziness was maximal. A similar behavior was found in Wanke, Barros, and Emrouznejad (2015) and Wanke, Barros, et al. (2016) when applying this bootstrapped regression to different decision-making contexts. This also suggests the use of different rule-based systems to assess the problem of predicting efficiency levels in BRICS banking.

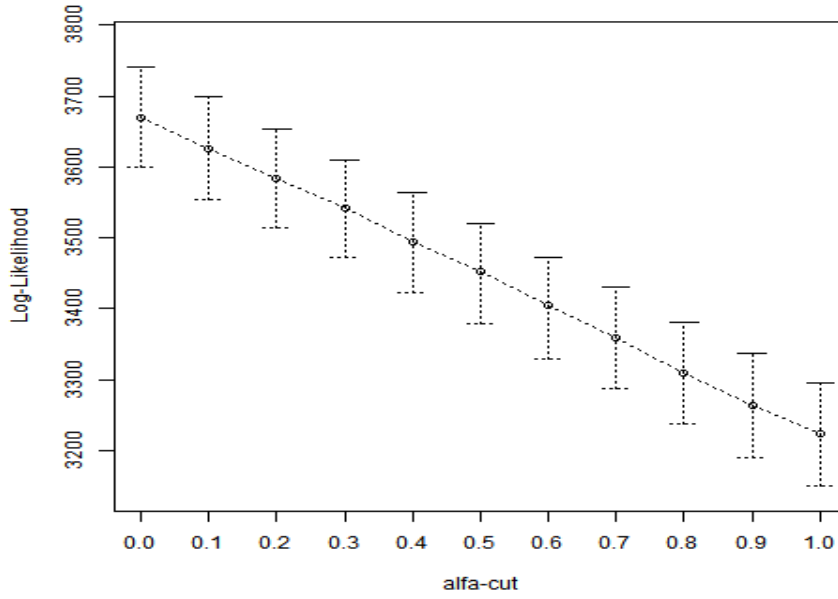
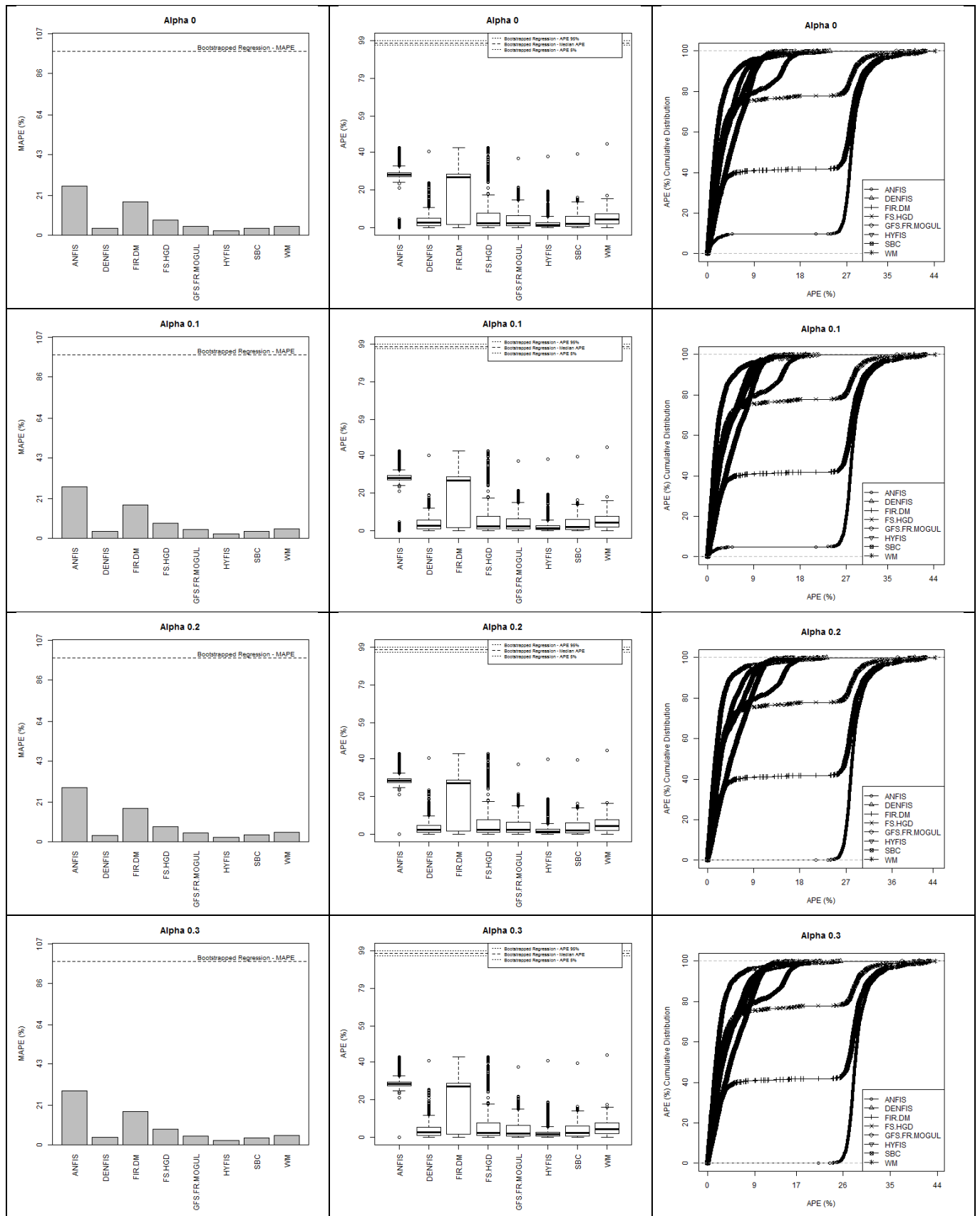
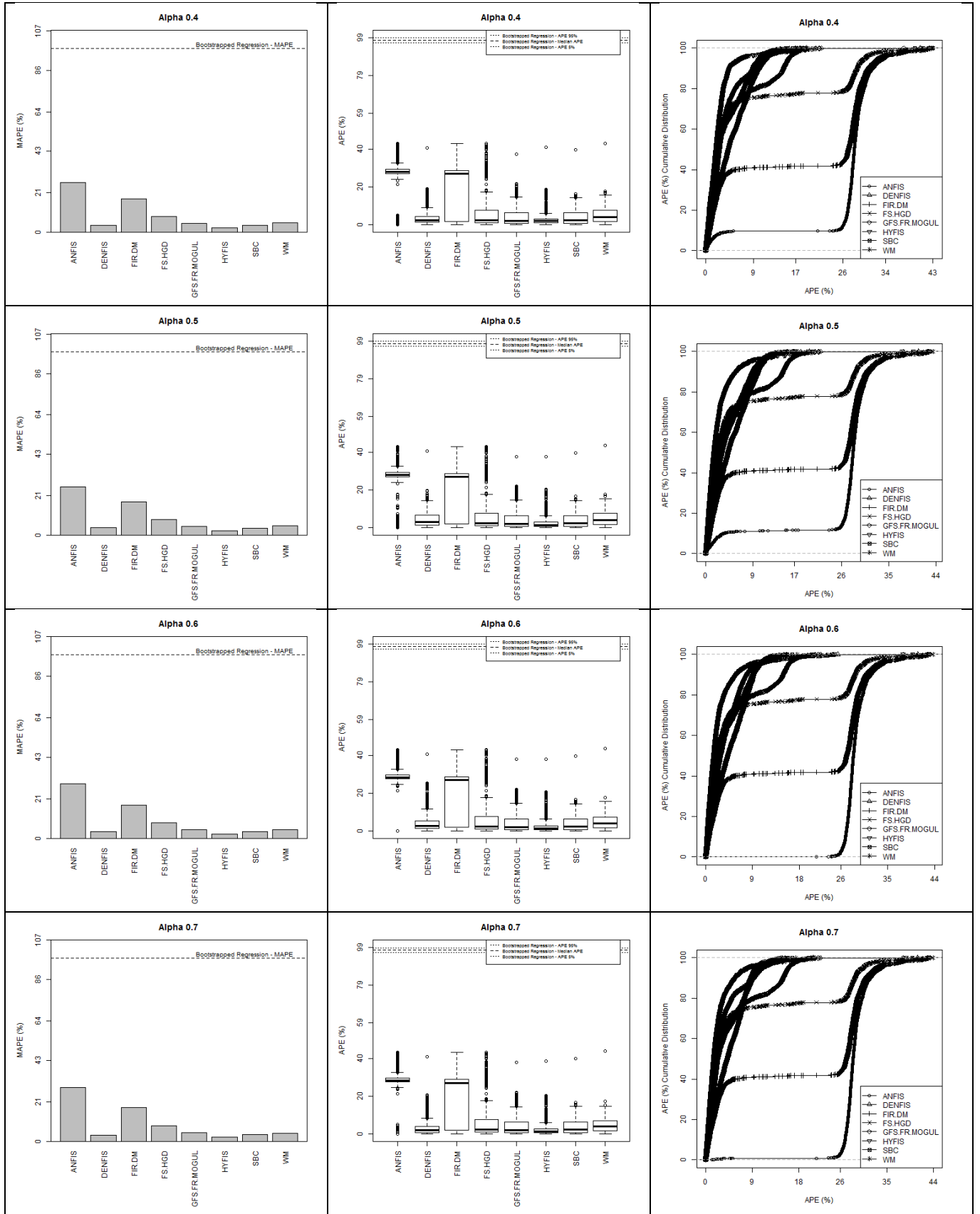


Fig. 3. Log-likelihood confidence intervals for the conditional bootstrap truncated regressions.

Therefore, the results of the rule-based systems methods presented in Tables 2 and 3 and considered for comparison within the ambit of fuzzy regressions are now discussed. To compare the results, we computed the APE – Average Percent Error. Two panels with the full set of results are shown in Figs. 4 and 5. One is organized by alpha-levels; the other, by methods. It can be seen that the eight FRBS methods tested outperformed bootstrapped truncated regression results in terms of MAPE. With respect to the distribution of the APEs, the average errors were substantially smaller under HYFIS, SBC, FS.HGD, DENFIS, GFS.FR.MOGUL, and WM models, and not even comparable in central tendency and distributional characteristics to those worse results obtained via bootstrap under ANFIS, and FIR.DM. This suggests that outperformance in terms of APE and MAPE can be achieved in several FRBS models, regardless of the underlying learning method. The absence of interaction between fuzziness and randomness, as detected in the bootstrapped regression, altogether with the fact that the best likelihood model is the one obtained for highest input/output fuzziness, may help in explaining why several FRBS methods performed superiorly best. Further research, however, is deemed necessary to confirm this conjecture under circumstances where there are interactions between randomness and fuzziness.





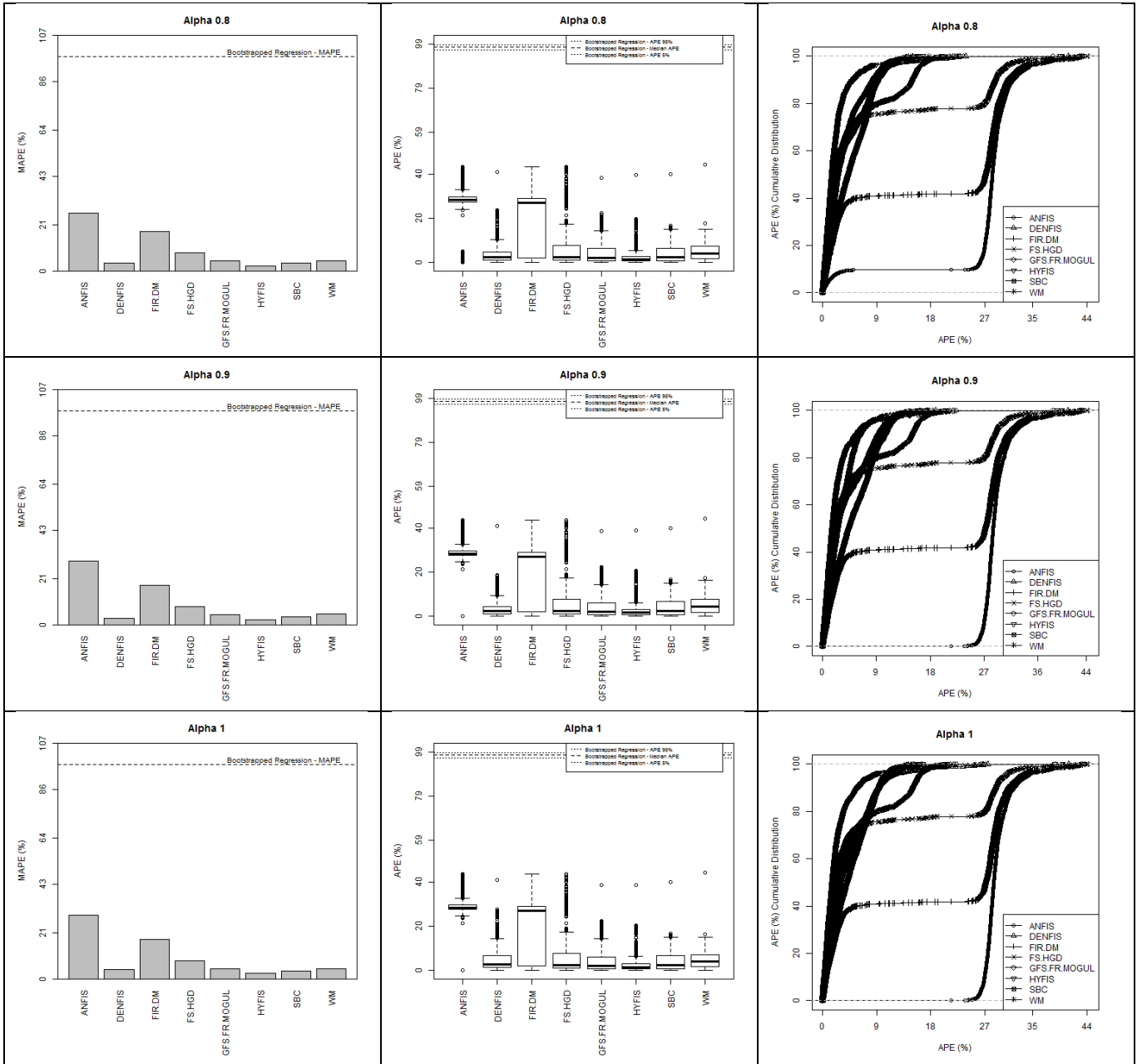
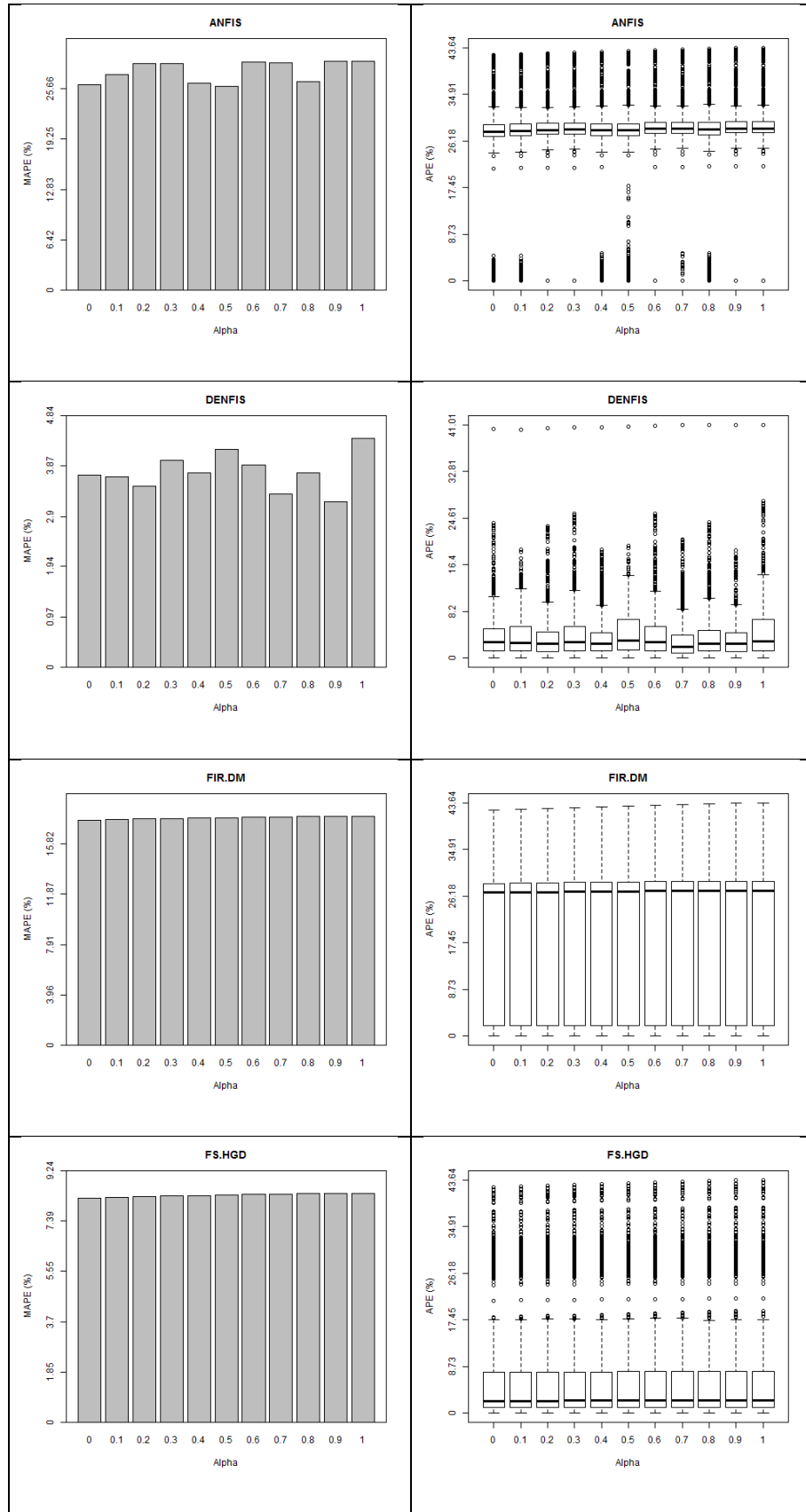


Fig. 4. FRBS regression results grouped by α -levels.

On the other hand, Fig. 5 illustrates the impact of the fuzziness level on the errors achieved by each FRBS model. It illustrates that the different alpha-levels do not impact much on the distributional aspect of the APE and its central tendency (median), although average errors (MAPE) tend to slightly increase with increasing fuzziness (alpha-level = 0) only under the HYFIS approach. Under the remainder approaches, either the APE increases or remains stagnant with lower fuzziness levels. This suggests that FRBS generally works better under higher fuzziness environments.



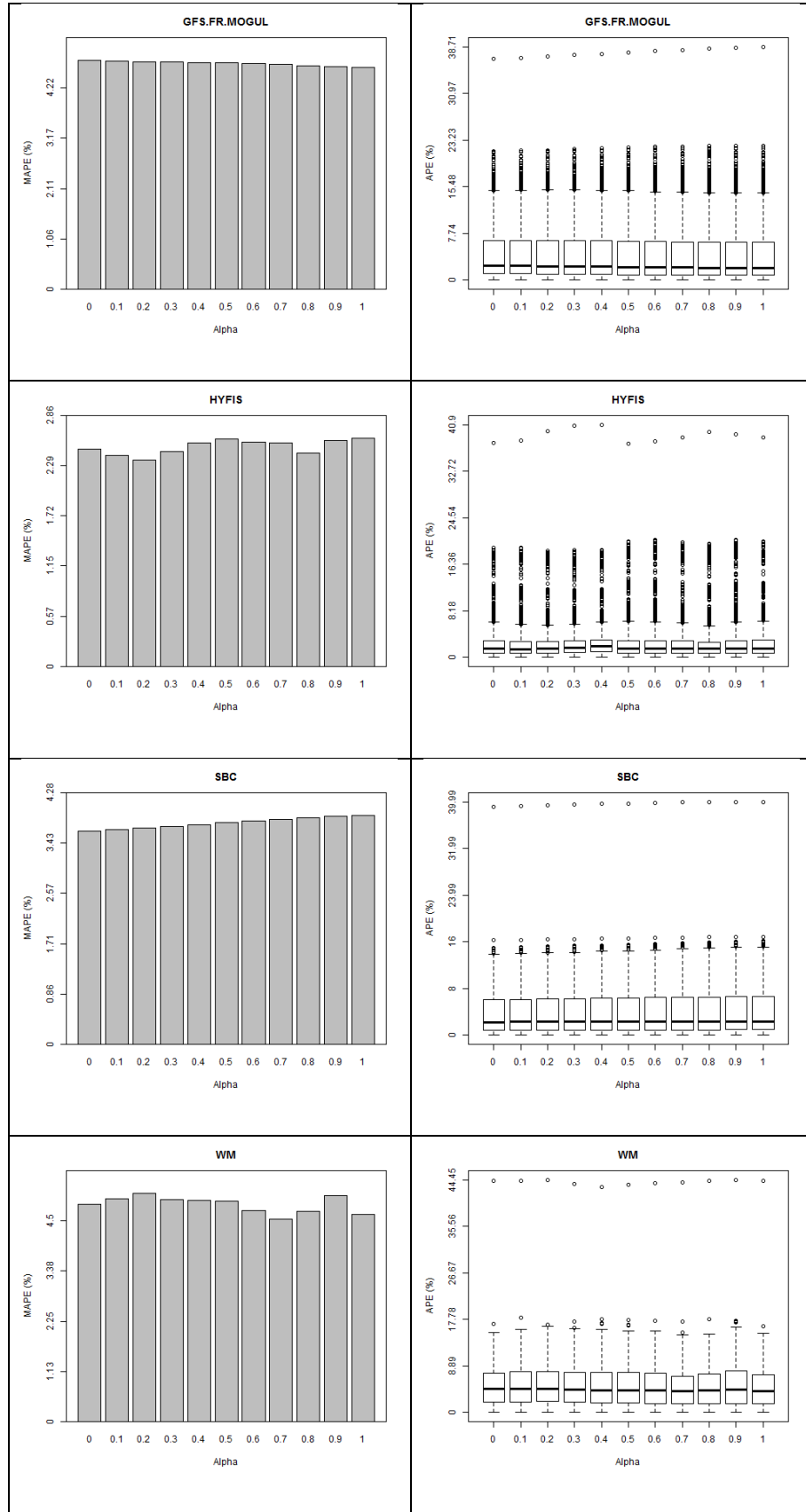


Fig. 5. FRBS regression results grouped by models.

The ruling implications for decision-makers based on this practical application in BRICS banks are related to very nature of the fuzziness and randomness of the problem under analysis. Considering that such interactions may exist, and can be detected in sign reversal for different alpha levels, one should consider using bootstrapped conditional regressions and FRBS regressions in a complimentary fashion. The first one should be used in detecting significant contextual variables; the other one, for prediction purposes. On the other hand, putting the case for the BRICS banks into perspective and considering that there was no interaction found between randomness and fuzziness embedded within the collected data, decision makers should adopt a combination approach, where different forecasts are weighted and combined into a final number.

5. Conclusion

This paper presents an analysis of BRICS banking efficiency using Fuzzy TOPSIS and fuzzy based regressions in a two-stage approach. Fuzzy TOPSIS enables different sources of uncertainty and vagueness to be handled while computing the efficiency scores. Based on the results of the fuzzy analysis performed in the second stage, it is possible to explain the causes of inefficiency in terms of socio-economic, regulatory, and demographic variables. Banking efficiency within the ambit of the BRICS countries appears to be explained by the countervailing forces of capital accumulation and social-welfare, building upon the trade-off between inflation ratio and country gross savings.

The contributions of this paper are as follows. Firstly, this paper explains and empirically presents the importance of using various regressions to describe bank efficiency by using fuzzy TOPSIS the most used method (Kahraman et al., 2015). The findings of this paper suggested that efficiency scores varies with respect to major contextual variables. So, in case of any event of national or international financial crisis, using profitability approach can better explain bank performance. Secondly, banks' heterogeneity is considered and the existing meta-frontier method is used with the above business approaches to measure bank efficiency. Third, Bootstrapping was also used to build confidence intervals for the log-likelihood measurements for the conditional regression models considering different α -values. It is interesting to note how the regression's performance is affected by the extreme values of uncertainty in the measurement of inputs and outputs.

From the managerial perspective, this paper contributes by examining bank efficiency using both the business approaches and contextual variables. This paper also serves as the ground breaking benchmarking tool to explain the diverse aspects of banking business. This paper suggests that for attaining bank efficiency managers should not only to focus on their peer banks but also to contextual variables of a bank before benchmarking.

References

- Amsler, C., Lee, Y. H., & Schmidt, P. (2009). A survey of stochastic frontier models and likely future developments. *Seoul Journal of Economics*, 22(1), 5-27.
- Asai, H. (1982). Linear Regression Analysis with Fuzzy Model. *IEEE Transactions on Systems, Man, and Cybernetics*, 12(6), 903-907. doi:<http://dx.doi.org/10.1109/TSMC.1982.4308925>
- Battese, G. E., & Rao, D. P. (2002). Technology gap, efficiency, and a stochastic metafrontier function. *International Journal of Business and Economics*, 1(2), 87-93.
- Behzadian, M., Khanmohammadi Otaghsara, S., Yazdani, M., & Ignatius, J. (2012). A state-of-the-art survey of TOPSIS applications. *Expert Systems with Applications*, 39(17), 13051-13069. doi:<http://dx.doi.org/10.1016/j.eswa.2012.05.056>
- Berger, A. N., & Humphrey, D. B. (1997). Efficiency of financial institutions: International survey and directions for future research. *European Journal of Operational Research*, 98(2), 175-212. doi:[http://dx.doi.org/10.1016/S0377-2217\(96\)00342-6](http://dx.doi.org/10.1016/S0377-2217(96)00342-6)
- Carlsson, C., & Fullér, R. (1996). Fuzzy Multiple Criteria Decision Making Fuzzy multiple criteria decision making: Recent developments. *Fuzzy Sets and Systems*, 78(2), 139-153. doi:[http://dx.doi.org/10.1016/0165-0114\(95\)00165-4](http://dx.doi.org/10.1016/0165-0114(95)00165-4)
- Chang, O. Y.-H., & Ayyub, B. M. (2001). Fuzzy regression methods – a comparative assessment. *Fuzzy Sets and Systems*, 119(2), 187-203. doi:[http://dx.doi.org/10.1016/S0165-0114\(99\)00091-3](http://dx.doi.org/10.1016/S0165-0114(99)00091-3)
- Chang, S.-H., & Tseng, H.-E. (2008). Fuzzy TOPSIS decision method for configuration management. *International Journal of Industrial Engineering: Theory, Applications and Practice*, 15(3), 304-313.
- Charnes, A., Cooper, W., & Rhodes, E. (1978). Measuring the efficiency of decision making units. *European Journal of Operational Research*, 2(6), 429-444. doi:[http://dx.doi.org/10.1016/0377-2217\(78\)90138-8](http://dx.doi.org/10.1016/0377-2217(78)90138-8)
- Chen, C.-T. (2000). Extensions of the TOPSIS for group decision-making under fuzzy environment. *Fuzzy Sets and Systems*, 114(1), 1-9. doi:[http://dx.doi.org/10.1016/S0165-0114\(97\)00377-1](http://dx.doi.org/10.1016/S0165-0114(97)00377-1)
- Cheng, H. F., Gutierrez, M., Mahajan, A., Shachmurove, Y., & Shahrokhi, M. (2007). A future global economy to be built by BRICs. *Global Finance Journal*, 18(2), 143-156. doi:<http://dx.doi.org/10.1016/j.gfj.2006.04.003>
- Choudhury, A. A., & Ashraf, E. (2016). Determinants of commercial banks' capital structure in BRIC. *International Journal of Entrepreneurship and Development Studies*, 4(1), 39-59.
- Demirgüç-Kunt, A., & Huizinga, H. (1999). Determinants of commercial bank interest margins and profitability: some international evidence. *The World Bank Economic Review*, 13(2), 379-408.
- Dubois, D. J. (1980). *Fuzzy sets and systems: theory and applications* (Vol. 144): Academic press.
- Dubois, D. J., & Prade, H. (1998). Possibility theory: qualitative and quantitative aspects *Quantified representation of uncertainty and imprecision* (pp. 169-226): Springer.
- Fethi, M. D., & Pasiouras, F. (2010). Assessing bank efficiency and performance with operational research and artificial intelligence techniques: A survey. *European Journal of Operational Research*, 204(2), 189-198. doi:<http://dx.doi.org/10.1016/j.ejor.2009.08.003>
- Hashimoto, H. (1983). Convergence of powers of a fuzzy transitive matrix. *Fuzzy Sets and Systems*, 9(1), 153-160. doi:[http://dx.doi.org/10.1016/S0165-0114\(83\)80015-3](http://dx.doi.org/10.1016/S0165-0114(83)80015-3)

- Jacobs, L. M., & Van Rossem, R. (2014). The BRIC Phantom: A comparative analysis of the BRICs as a category of rising powers. *Journal of Policy Modeling*, 36, Supplement 1, S47-S66.
doi:<http://dx.doi.org/10.1016/j.jpolmod.2013.10.008>
- Jefmański, B., & Dudek, A. (2015). The fuzzy TOPSIS method and its implementation in the R programme. *Informatyka Ekonomiczna*(1 (35)), 19-27.
- Kahraman, C., Onar, S. C., & Oztaysi, B. (2015). Fuzzy Multicriteria Decision-Making: A Literature Review. *International Journal of Computational Intelligence Systems*, 8(4), 637-666.
doi:<http://dx.doi.org/10.1080/18756891.2015.1046325>
- Kandel, A. (1986). Fuzzy mathematical techniques with applications. Reading: Addison-Wesley, 1986.
- Kia, S., Danaei, A., & Oroei, M. (2014). An application of fuzzy TOPSIS on ranking products: A case study of faucet devices. *Decision Science Letters*, 3(1), 43-48.
- Klir, G., & Yuan, B. (1995). *Fuzzy sets and fuzzy logic* (Vol. 4): Prentice hall New Jersey.
- Laïdi, Z. (2012). BRICS: Sovereignty power and weakness. *International Politics*, 49(5), 614-632.
doi:<http://dx.doi.org/10.1057/ip.2012.17>
- Liu, J. S., Lu, L. Y. Y., Lu, W.-M., & Lin, B. J. Y. (2013a). Data envelopment analysis 1978–2010: A citation-based literature survey. *Omega-International Journal of Management Science*, 41(1), 3-15.
doi:<http://dx.doi.org/10.1016/j.omega.2010.12.006>
- Liu, J. S., Lu, L. Y. Y., Lu, W.-M., & Lin, B. J. Y. (2013b). A survey of DEA applications. *Omega-International Journal of Management Science*, 41(5), 893-902.
doi:<http://dx.doi.org/10.1016/j.omega.2012.11.004>
- Luminita, C., Constantin, C., Valeriu, I.-F., Radulescu, I. G., Panait, M., & Voica, C. (2014). 1st International Conference 'Economic Scientific Research - Theoretical, Empirical and Practical Approaches', ESPERA 2013BRICS Countries Challenge to the World Economy New Trends. *Procedia Economics and Finance*, 8, 605-613. doi:[http://dx.doi.org/10.1016/S2212-5671\(14\)00135-X](http://dx.doi.org/10.1016/S2212-5671(14)00135-X)
- Luna, V. M. I. (2016). brics' bank: possibilities and constraints. *Economía Informa*, 398, 3-22.
doi:<http://dx.doi.org/10.1016/j.ecin.2016.04.002>
- Mobarek, A., & Fiorante, A. (2014). The prospects of BRIC countries: Testing weak-form market efficiency. *Research in International Business and Finance*, 30, 217-232.
doi:<http://dx.doi.org/10.1016/j.ribaf.2013.06.004>
- O'Donnell, C. J., Rao, D. S. P., & Battese, G. E. (2008). Metafrontier frameworks for the study of firm-level efficiencies and technology ratios. *Empirical Economics*, 34(2), 231-255.
doi:<http://dx.doi.org/10.1007/s00181-007-0119-4>
- O'Neill, J. (2001). *Building better global economic BRICs*. Retrieved from
- O'Neill, J. (2011). *The Growth Map: Economic opportunity in the BRICs and beyond*: Penguin UK.
- Paradi, J. C., & Zhu, H. Y. (2013). A survey on bank branch efficiency and performance research with data envelopment analysis. *Omega-International Journal of Management Science*, 41(1), 61-79.
doi:<http://dx.doi.org/10.1016/j.omega.2011.08.010>
- Pedrycz, W., & Gomide, F. (1998). *An introduction to fuzzy sets: analysis and design*: Mit Press.
- Porcelli, F. (2009). Measurement of Technical Efficiency. A brief survey on parametric and non-parametric techniques. *University of Warwick*, 11.
- Riza, L. S., Bergmeir, C. N., Herrera, F., & Benítez Sánchez, J. M. (2015). *frbs: fuzzy rule-based systems for classification and regression in R*.
- Sampaio, A. (2013). Review of Frontier Models and Efficiency Analysis: A Parametric Approach. In B. A. Mendes, E. L. D. G. Soares da Silva, & M. J. Azevedo Santos (Eds.), *Efficiency Measures in the Agricultural Sector: With Applications* (pp. 13-35). Dordrecht: Springer Netherlands.
- Simar, L., & Wilson, P. W. (2007). Estimation and inference in two-stage, semi-parametric models of production processes. *Journal of Econometrics*, 136(1), 31-64.
doi:<http://dx.doi.org/10.1016/j.jeconom.2005.07.009>
- Syamal, A. K., & Pal, M. (2007). Triangular fuzzy matrices. *Iranian Journal of Fuzzy Systems*, 4(1), 75-87.
- Teh, C. S., Chae, H.-R., Aduce, S. A. Z., Anding, P. N., Chen, C. J., Aziz, N. A., . . . Abdullah, L. (2013). The 9th International Conference on Cognitive Science Fuzzy Multi Criteria Decision Making and its

- Applications: A Brief Review of Category. *Procedia - Social and Behavioral Sciences*, 97, 131-136. doi:<http://dx.doi.org/10.1016/j.sbspro.2013.10.213>
- Thomason, M. G. (1977). Convergence of powers of a fuzzy matrix. *Journal of Mathematical Analysis and Applications*, 57(2), 476-480. doi:[http://dx.doi.org/10.1016/0022-247X\(77\)90274-8](http://dx.doi.org/10.1016/0022-247X(77)90274-8)
- Uyun, S., & Riadi, I. (2013). A fuzzy TOPSIS multiple-attribute decision making for scholarship selection. *arXiv preprint arXiv:1306.6489*.
- Wanke, P., Azad, M. A. K., & Barros, C. P. (2016). Financial distress and the Malaysian dual banking system: A dynamic slacks approach. *Journal of Banking & Finance*, 66, 1-18. doi:<http://dx.doi.org/10.1016/j.jbankfin.2016.01.006>
- Wanke, P., Azad, M. A. K., Barros, C. P., & Hadi-Vencheh, A. (2015). Predicting performance in ASEAN banks: an integrated fuzzy MCDM–neural network approach. *Expert Systems*, Article first published online: 18 DEC 2015. doi:<http://dx.doi.org/10.1111/exsy.12144>
- Wanke, P., Barros, C. P., & Emrouznejad, A. (2015). Assessing Productive Efficiency of Banks Using Integrated Fuzzy-DEA and Bootstrapping: A Case of Mozambican Banks. *European Journal of Operational Research*, 249(1), 378-389. doi:<http://dx.doi.org/10.1016/j.ejor.2015.10.018>
- Wanke, P., Barros, C. P., & Faria, J. R. (2015). Financial distress drivers in Brazilian banks: A dynamic slacks approach. *European Journal of Operational Research*, 240(1), 258-268. doi:<http://dx.doi.org/10.1016/j.ejor.2014.06.044>
- Wanke, P., Barros, C. P., & Nwaogbe, O. R. (2016). Assessing productive efficiency in Nigerian airports using Fuzzy-DEA. *Transport Policy*, 49, 9-19. doi:<http://dx.doi.org/10.1016/j.tranpol.2016.03.012>
- Yayla, A. Y., Yildiz, A., & Ozbek, A. (2012). Fuzzy TOPSIS method in supplier selection and application in the garment industry. *Fibres & Textiles in Eastern Europe*.
- Zadeh, L. A. (1965). Fuzzy sets. *Information and Control*, 8(3), 338-353. doi:[http://dx.doi.org/10.1016/S0019-9958\(65\)90241-X](http://dx.doi.org/10.1016/S0019-9958(65)90241-X)
- Zhang, J., Jiang, C., Qu, B., & Wang, P. (2013). Market concentration, risk-taking, and bank performance: Evidence from emerging economies. *International Review of Financial Analysis*, 30, 149-157. doi:<http://dx.doi.org/10.1016/j.irfa.2013.07.016>
- Zhou, P., Ang, B. W., & Poh, K. L. (2008). A survey of data envelopment analysis in energy and environmental studies. *European Journal of Operational Research*, 189(1), 1-18. doi:<http://dx.doi.org/10.1016/j.ejor.2007.04.042>